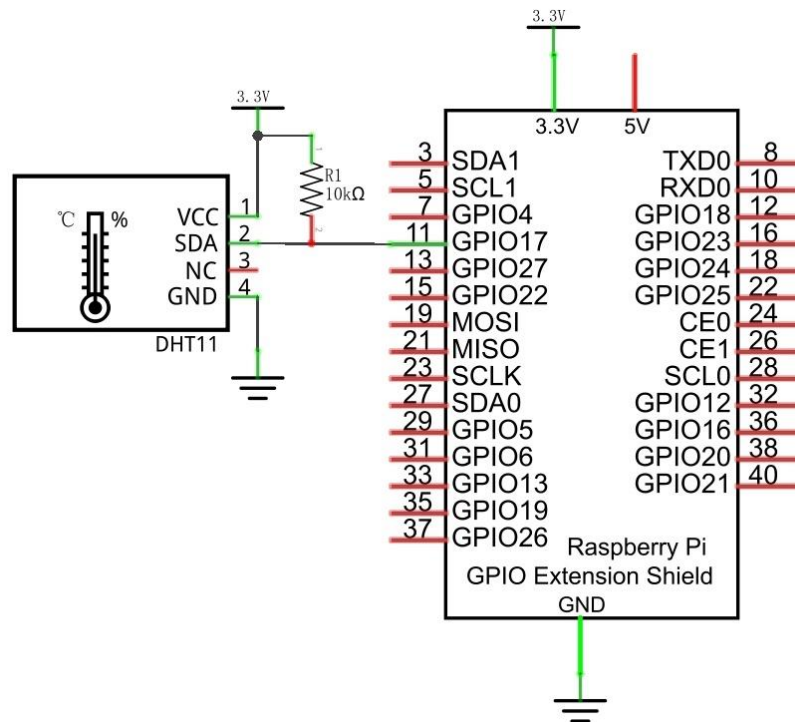
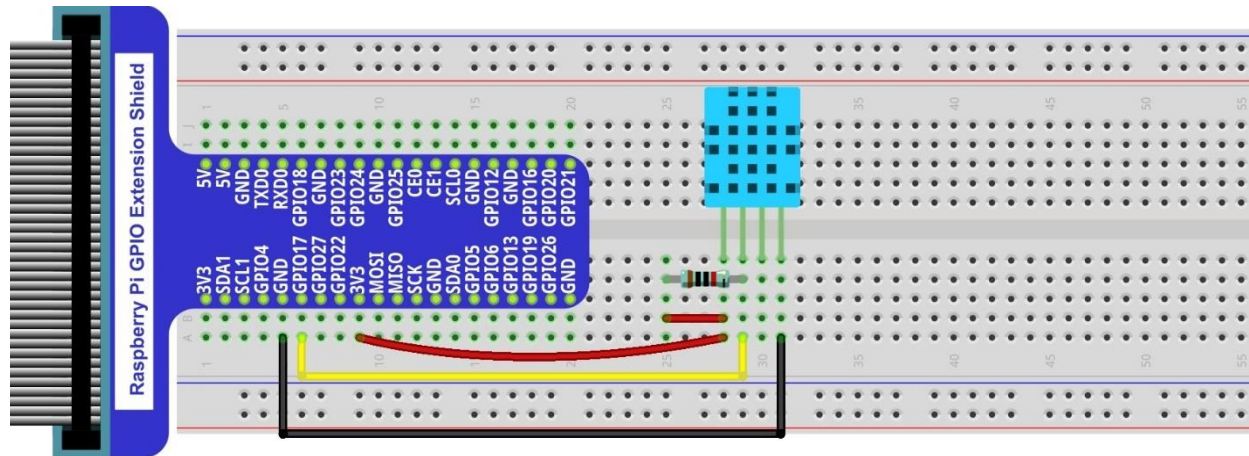


Circuit

Schematic diagram



Hardware connection. If you need any support, please feel free to contact us via: support@freenove.com



Video: <https://www.youtube.com/watch?v=vTeuLWiXUTA>

Code

The code is used to read the temperature and humidity data of DHT11, and display them.

Python Code 21.1.1 DHT11

First, observe the project result, and then learn about the code in detail.

If you have any concerns, please contact us via: support@freenove.com

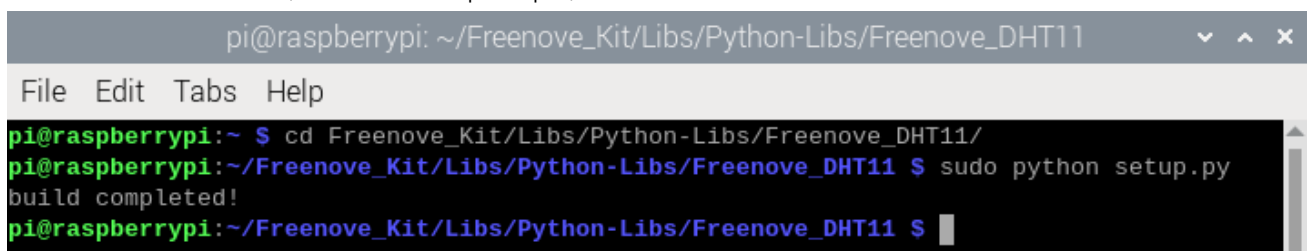
1. Use cd command to enter folder of the ADC Device library.

```
cd ~/Freenove_Kit/Libs/Python-Libs/Freenove_DHT11
```

2. Execute command below to install the library.

```
sudo python setup.py
```

A successful installation, without error prompts, is shown below:



```
pi@raspberrypi: ~/Freenove_Kit/Libs/Python-Libs/Freenove_DHT11
File Edit Tabs Help
pi@raspberrypi:~ $ cd Freenove_Kit/Libs/Python-Libs/Freenove_DHT11/
pi@raspberrypi:~/Freenove_Kit/Libs/Python-Libs/Freenove_DHT11 $ sudo python setup.py
build completed!
pi@raspberrypi:~/Freenove_Kit/Libs/Python-Libs/Freenove_DHT11 $
```

Next, we will execute the code for this project.

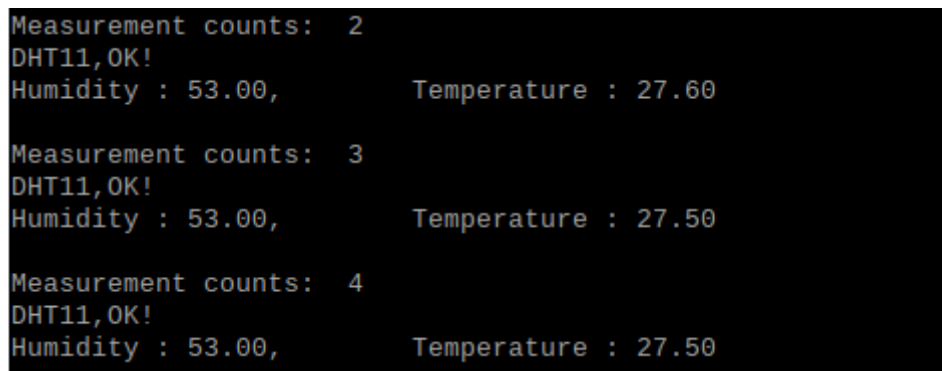
1. Use cd command to enter 21.1.1_DHT11 directory of Python code.

```
cd ~/Freenove_Kit/Code/Python_GPIOZero_Code/21.1.1_DHT11
```

2. Use Python command to execute code "DHT11.py".

```
python DHT11.py
```

After the program is executed, the Terminal window will display the current total number of read times, the read state, as well as temperature and humidity values as is shown below:



```
Measurement counts: 2
DHT11,OK!
Humidity : 53.00,      Temperature : 27.60

Measurement counts: 3
DHT11,OK!
Humidity : 53.00,      Temperature : 27.50

Measurement counts: 4
DHT11,OK!
Humidity : 53.00,      Temperature : 27.50
```

Since gpiozero does not support DHT11 sensors, RPi.GPIO is used here for control.

The following is the program code:

```
1 import time
2 import Freenove_DHT as DHT
3 DHTPin = 17      #define the pin of DHT11
4
5 def loop():
6     dht = DHT.DHT(DHTPin)  #create a DHT class object
7     counts = 0 # Measurement counts
```

```

8      while(True):
9          counts += 1
10         print("Measurement counts: ", counts)
11         for i in range(0,15):
12             chk = dht.readDHT11()      #read DHT11 and get a return value. Then determine
whether data read is normal according to the return value.
13             if (chk is dht.DHTLIB_OK):      #read DHT11 and get a return value. Then determine
whether data read is normal according to the return value.
14                 print("DHT11, OK!")
15                 break
16             time.sleep(0.1)
17             print("Humidity : %.2f, \t Temperature : %.2f \n"%( dht.getHumidity(),
dht.getTemperature()))
18             time.sleep(2)
19
20 if __name__ == '__main__':
21     print ('Program is starting ... ')
22     try:
23         loop()
24     except KeyboardInterrupt:
25         exit()

```

In this project code, we use a module "**Freenove_DHT.py**", which provides the method of reading the DHT Sensor. It is located in the same directory with program files "**DHT11.py**". By using this library, we can easily read the DHT Sensor. First, we create a DHT class object in the code.

```
dht = DHT.DHT(DHTPin)      #create a DHT class object
```

Then in the "while" loop, use `chk = dht.readDHT11()` to read the DHT11, and determine whether the data read is normal according to the return value "chk". Then use variable sumCnt to record the number of times read.

```

while(True):
    counts += 1
    print("Measurement counts: ", counts)
    for i in range(0,15):
        chk = dht.readDHT11()      #read DHT11 and get a return value. Then determine
whether data read is normal according to the return value.
        if (chk is dht.DHTLIB_OK):      #read DHT11 and get a return value. Then determine
whether data read is normal according to the return value.
            print("DHT11, OK!")
            break
        time.sleep(0.1)
        print("Humidity : %.2f, \t Temperature : %.2f \n"%( dht.getHumidity(),
dht.getTemperature()))
        time.sleep(2)

```

Finally display the results:

```
print("Humidity : %.2f, \t Temperature : %.2f \n"%( dht.getHumidity(), dht.getTemperature()))
```

Module "**Freenove_DHT.py**" contains a DHT class. The class function of the def **readDHT11** (pin) is used to read the DHT11 Sensor and store the temperature and humidity data read to member variables humidity and temperature.

Freenove_DHT Module

This is a Python module for reading the temperature and humidity data of the DHT Sensor. Partial functions and variables are described as follows:

getHumidity(): store humidity data read from sensor

getTemperature(): store temperature data read from sensor

readDHT11(): read the temperature and humidity of sensor DHT11, and return values used to determine whether the data is normal.